Blockchain Justification

Purpose of Using Blockchain

The reputation system in this project is implemented as a smart contract on Ethereum. Blockchain is used to provide:

- Deterministic rule execution
- Immutable record of updates
- Transparent auditability
- Decentralized state management

The goal is not to use blockchain for dead hype but to leverage its structural properties.

Deterministic Execution

Ethereum smart contracts execute deterministically. Meaning:

- The same input always produces the same output.
- All nodes in the network reach the same result.
- The contract logic cannot behave differently for different users.

For a reputation system, this ensures that:

- Reputation updates are predictable.
- Decay rules are consistently applied.
- Explanation outputs are reproducible.

Immutability

Blockchain stores data in blocks that are cryptographically linked. Once a transaction is confirmed:

- It cannot be altered.
- It cannot be deleted.
- Its history remains publicly verifiable.

Here, immutability ensures:

- Reputation changes cannot be secretly modified.
- Explanation logs cannot be tampered with.
- The update history remains transparent.

Auditability

All smart contract events are recorded on-chain, ensuring

- Anyone to verify reputation updates.
- External tools to read explanation logs.
- Independent validation of system behavior.

Auditability is especially important for reputation systems, where fairness and transparency are critical. To avoid systemic biasness.

Time Measurement

Blockchain provides access to block.timestamp, which allows the contract to measure elapsed time between interactions. This enables implementation of time-based decay without requiring an external scheduler.

Although timestamps are approximate, they are sufficient for coarse-grained intervals such as days or weeks.

Why Not a Centralized Database?

A centralized database could implement similar logic. But →

- The system operator could modify scores.
- Explanation logs could be altered.
- Users would have to trust a central authority.

By deploying the logic on Ethereum →

- The rules become transparent.
- The execution becomes verifiable.
- The update history becomes immutable.

This aligns with the objective of creating a transparent and explainable reputation mechanism.

Design Philosophy

Blockchain is used here as a deterministic execution layer, transparent ledger and as an immutable audit log. The design intentionally keeps blockchain usage minimal and focused.

It is not used for →

- Token issuance,
- Financial incentives,
- Complex governance mechanisms.